General Information

- Download and read the paper

- **A Survey of Smart Parking Solutions**
- IEEE Transactions on Intelligent Transportation Systems, vol. 18, no. 12, pp. 3229-3253.

- https://ieeexplore.ieee.org/document/7895130

- (or with NTU Login)
- https://ieeexplore.ieee.org.remotexs.ntu.edu.sg/document/7895130

Video Demos

- Basic demo: https://www.youtube.com/watch?v=-9s9QkpRzWs

- Singapore barrier-less parking https://www.youtube.com/watch?v=bsW9s1AIQ00

Review of the Survey Paper

- Essence of a smart parking system
  - Flow of traffic
    - Vehicles moves on streets looking for parking spaces
    - Vehicles flow into parking lots
    - Vehicles flow out from parking lots
  - Flow of information
    - Vacancy information from field sensors to information system
    - Information system sends vacancy/price information to vehicles
  - Cyber intelligence
    - Processes requests from vehicles

Review of the Survey Paper (cont'd)

- Three main areas of study
  - Information collection
    To identify parking spaces status
  - System deployment
    Software system exploitation and statistical analysis of collected data
  - Service dissemination
    How to provide the service to vehicles (e.g., how to solve competition)

Review of the Survey Paper (cont'd)

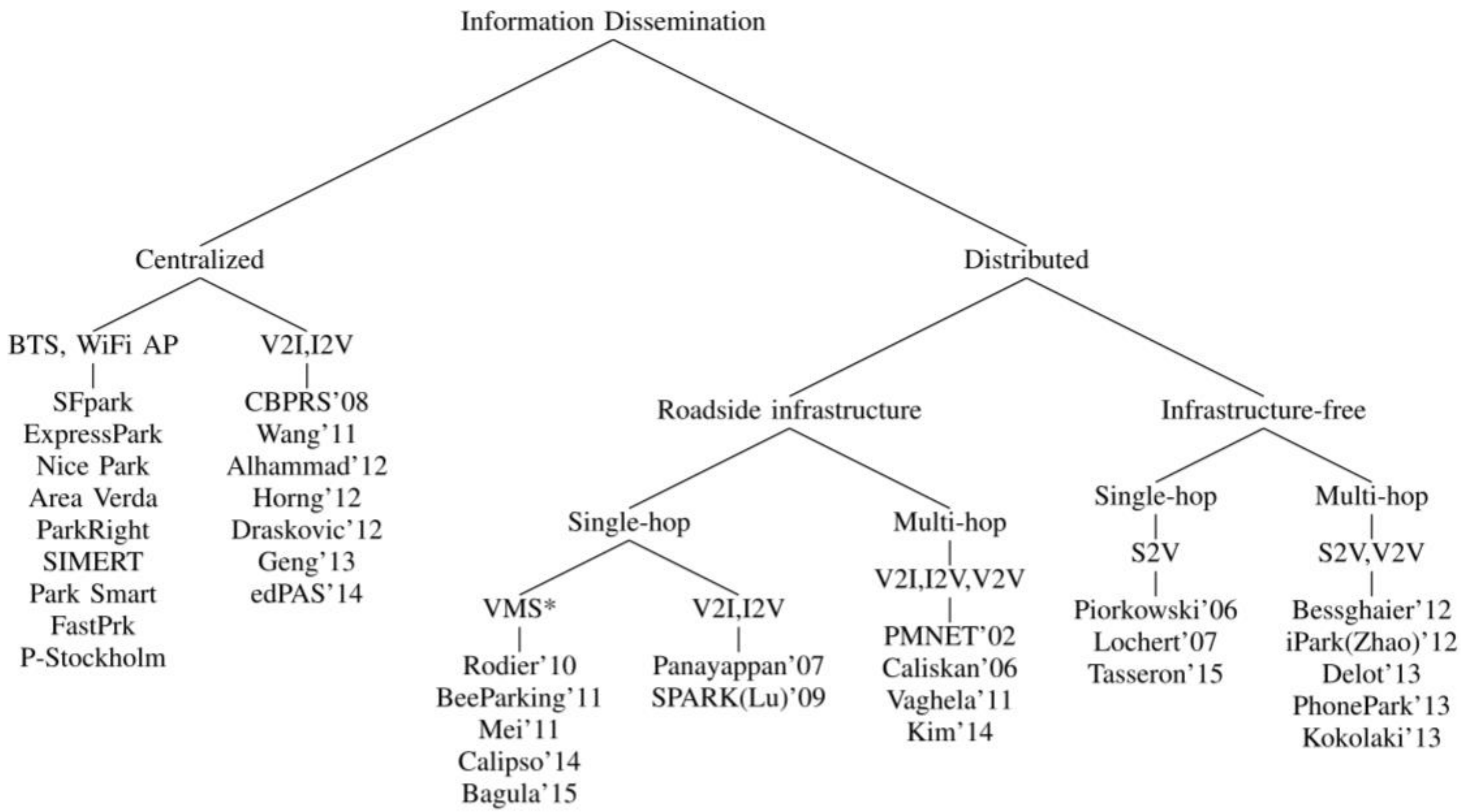
- Information sensing
  - Sensors
    - Passive infrared, ultrasonic, accelerometer, optical, magnetic, camera, acoustic, inductive loop, piezoelectric, RFID, laser rangefinder, robot
  - Sensor connectivity
    - Short-range, long-range
    - Wireless sensor networks
  - Parking meter
  - Crowdsensing
  - Shared parking

Review of the Survey Paper (cont'd)

- System deployment
  - Software system
    - Information management for sensor data
    - E-parking service for communicating with users
    - Software solution to handle simultaneous requests
    - User interface
    - Guidance
  - Large-scale deployment
    - ParkNet, San Francisco
    - Sfpark, San Francisco
    - LA ExpressPark, New York City
    - Nice Park, Nice
    - Etc
  - Parking vacancy prediction

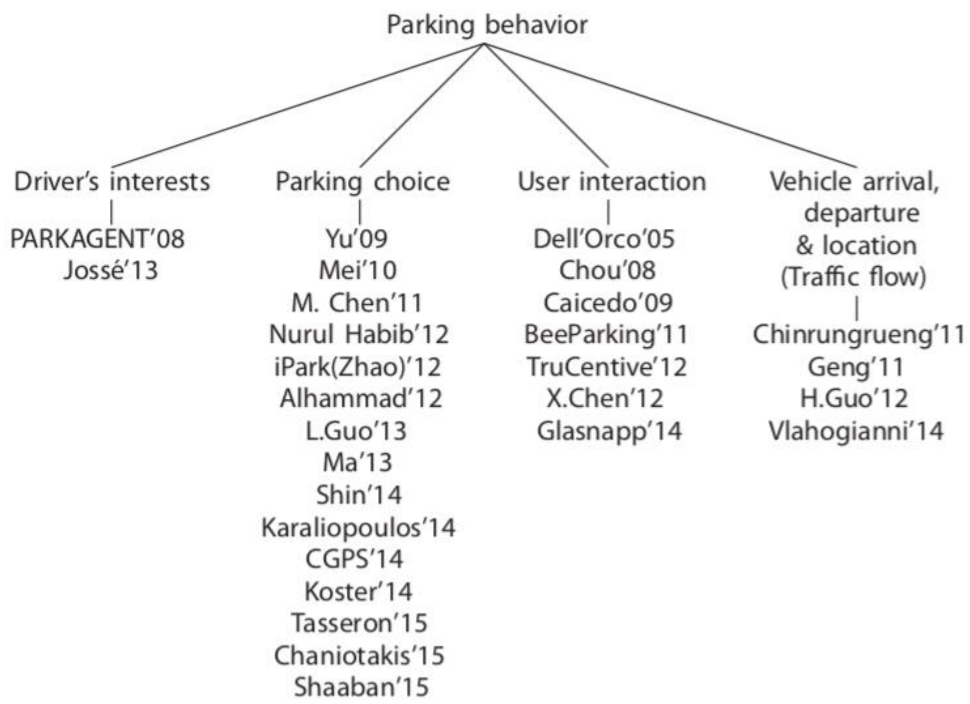
Review of the Survey Paper (cont'd)

- Parking service dissemination
  - Information dissemination



Review of the Survey Paper (cont'd)

- Parking competition
  - Dynamic pricing based on congestion status
  - Optimal pricing based on service provider's revenue
- Driver's parking behavior



Question 1

State the goal and benefits of Smart Parking applications

- <u>Goal</u>
- Smart parking is a way to help drivers find more efficiently satisfying parking spaces through information and communications technology
- <u>Benefits</u>
- Drivers can shorten their parking search time, reduce environmental pollution, reduce costs with less fuel consumption and alleviate traffic congestion through information from smart parking apps and also increases public transportation use rate and cities' revenues
- If drivers can rapidly find a parking space, the idle time for on-street parking is shorter and the parking revenues increase
- Installing sensors on unauthorized areas where people frequently park their car can help detect illegal parking and can issue in a penalty charge, as with electric vehicles and disabled parking stalls
- Once the traffic is fluent, it increases urban mobility and expands cities' capacities. Consequently, it brings more population, activities and business opportunities.

Question 2

Give an example scenario (a sequence of actions) of smart parking functionality

- Smart Parking smartphone application sends a request for parking demand from driver as well as the current and destination locations of the car
- Parking provider sends current availability and price of parking space to the driver
  - Parking provider recommends parking space (e.g., on street)
- The driver accepts and makes a reservation of available parking space
- The driver arrives at the parking space and makes payment
- Parking provider processes payment transaction

Question 3

List 6 sensors that can be used in Smart Parking solutions, explain each of the sensors briefly and its benefit

- Passive infrared sensors receive heat radiated from the human body and are often used collaboratively with other sensors to know if a driver parks and gets off his/her car.
- Cameras and acoustic sensors provide a much more complicated signal pattern than ultrasonic. Both of them require image and sonar processing in order to extract the desired information from the background noise.
- Inductive loops and piezoelectric sensors are both contactive and installed on road surface for traffic surveillance. They detect if a vehicle is passing. Piezoelectric sensors are similar to inductive loops but able to read more detailed information from the pressure exerted on it.

Question 3

List 6 sensors that can be used in Smart Parking solutions, explain each of the sensors briefly and its benefit

- RFID tags can be detected by RFID readers installed on parking spaces.

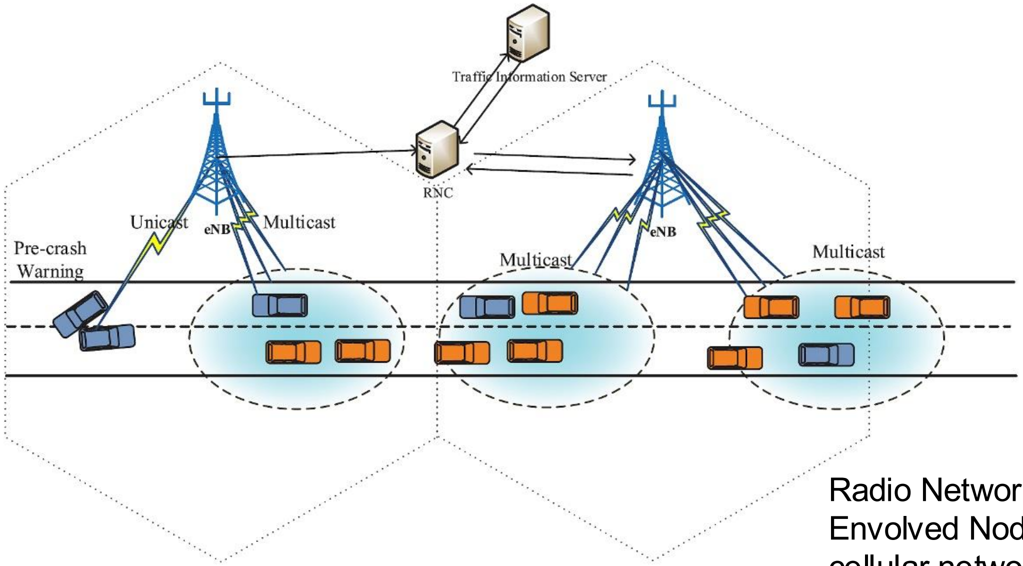
- Laser rangefinder is used to build a 1/2/3D map, especially for environmental perception. Normally installed on vehicles, it emits a laser beam and calculates the time of flight to measure the distances from different objects in order to know if there are parked vehicles

- Intelligent robots cruising on the street can also be used to recognize available parking spaces.

Question 4

Explain the data dissemination through communications and network infrastructure for Smart Parking applications

- Vehicle-to-Infrastructure (V2I) communications provide a connection between vehicles and the infrastructure located on the roadside such as base stations and access points, called RoadSide Unit (RSU)
- Infrastructure can be cellular networks and IEEE 802.11p/1609 or Dedicated Short Range Communications (DSRC)



Radio Network Controller (RNC)
Envolved NodeB (eNB), base station in LTE cellular networks

K. Zheng, Q. Zheng, P. Chatzimisios, W. Xiang and Y. Zhou, "Heterogeneous Vehicular Networking: A Survey on Architecture, Challenges, and Solutions," in IEEE Communications Surveys & Tutorials, vol. 17, no. 4, pp. 2377-2396, Fourthquarter 2015.

Question 5

Explain the concept of Crowdsensing and its role(s) in Smart Parking, state its limitations and/or shortcomings

- Crowdsensing is a method that allows a large population of users having different types of sensors, e.g., with their smartphones tablet, and wearable devices, to collect and share data with other entities. The data can be used to extract useful information to measure, analyze, estimate or predict, and optimize the systems and applications such as Smart Parking

- Smart Parking benefits from crowdsensing in which users can collect information about parking space availability and occupancy; some examples
  - Accelerometer and GPS are used to detect users' movements and parking or unparking status of a user
  - WiFi signal is detected if user is back to the car or if he/she is going from/to car
  - A detection method using the 3D compass of drivers' smartphone can be introduced. Thus, smartphones can detect if drivers are parking
  - Crowdsensing can allow users to share their parking space

Question 5

- Crowdsensing in Smart Parking has limitations include
  - Information accuracy: Sensors in smartphone may not provide accurate information, e.g., GPS data;
  - Trustless data: Users can manipulate sensor and parking data, e.g., indicating falsely parking space is occupied or free
  - Participation rates: Not many users may contribute parking data from sensors as collecting and transmitting it consumes energy from battery, e.g., no enough data results in inaccurate prediction
  - Freeriders: Some users never share data, but use data from other users

Question 6

List and explain 5 attributes (information) that Smart Parking applications should predict

- Parking vacancy
  - To inform drivers of possibility of getting parking space at certain locations and time
  - Useful for finding parking space, especially far from destination
- Parking occupancy rate
  - To inform drivers of possibility of arriving at the destination without parking space
- Vehicle population
  - Indication of traffic jam if parking space needs to be found
- Parking price
  - How much drivers has to pay by parking at certain location and time
- Origin-destination trip time
  - How much time required by drivers to reach parking space

Question 7

Consider exponential smoothing prediction model which is described by the forecast equation and smoothing equation defined as follows:

Forecast equation	**y(t+1|t)**	**=**	**l(t)**
Smoothing equation	**l(t)**	**=**	**a*y(t) + (1-a)*l(t-1)**

where l(t) is the smoothed value of the series at time t, and a is the smoothing parameter.

Calculate the car park occupancy rate at time 12noon from the occupancy rates given in the table and a = 0.5

Time	Car park occupancy rate
9:00am	80%
9:30am	82%
10:00am	85%
10:30am	83%
11:00am	90%
11:30am	92%

Question 7

- We have

I(9:00)	=	y(9:00)	=	80%
I(9:30)	=	0.5*y(9:30) + 0.5*I(9:00)	=	81%
I(10:00)	=	0.5*y(10:00) + 0.5*I(9:30)	=	83%
I(10:30)	=	0.5*y(10:30) + 0.5*I(10:00)	=	83%
I(11:00)	=	0.5*y(11:00) + 0.5*I(10:30)	=	86.5%
I(11:30)	=	0.5*y(11:30) + 0.5*I(11:00)	=	89.25%

y(12:00) = I(11:30) = **89.25%**

Time	Car park occupancy rate
9:00am	80%
9:30am	82%
10:00am	85%
10:30am	83%
11:00am	90%
11:30am	92%

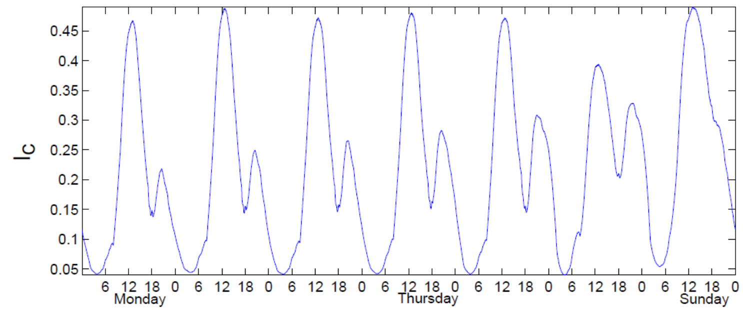
Question 8

Explain competition in Smart Parking, analyze and discuss one solution to resolve the competition

- Reason
- Parking competition always happens when more than two drivers contend the same parking space
- Parking competition can occur among parking providers for sharing parking demand
- Solutions
- Dynamic pricing is to regulate parking occupancy status and traffic congestion
- Optimal pricing to ensure maximum profit for parking providers given demand for parking

Question 8

- *Dynamic Pricing*
- Adjusting price based on the average occupancy of car park in a certain period



- Avoid underuse and congestion
- Underuse: possible users are discouraged and spaces remain unused that would be used at a lower rate
- Congestion: parkers with a high valuation are blocked, drivers circling for a space congest and pollute

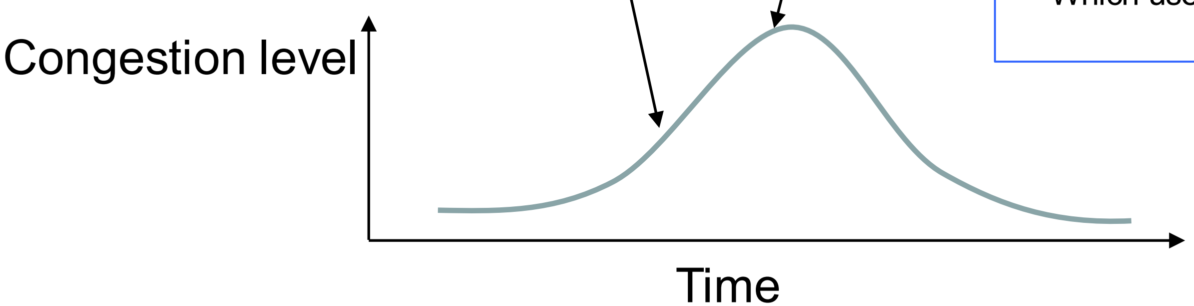
O. Zoeter, C. Dance, S. Clinchant, and J. Andreoli, "New algorithms for parking demand management and a city-scale deployment," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2014, pp. 1819–1828.

Question 8

- *Dynamic Pricing*
- Analysis:
- If rates are changed based on average occupancy, a period of underuse followed by congestion can lead to a perfect “neither too empty, nor too full” (Goldilocks) situation

Users search for parking space and price
- Price is not so expensive as congestion is not very heavy
- Many users go there to park

Users arrive and experience congestion

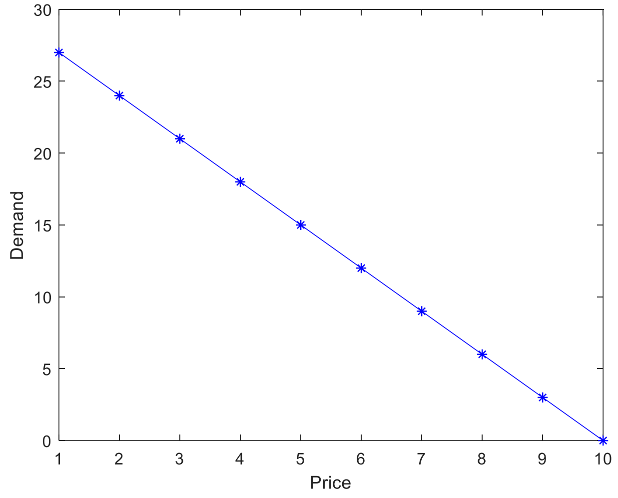


Use prediction to determine price
- How much time that users will reach car park?
- Which user to prioritize (nearest?)

Question 9

Optimal pricing:

The demand of a car park in one hour is described by a function D(p) = 30-3p, where D(p) is the number of cars wanted to park at the car park if the price for each parked car is p. Basically, the demand decreases as the price increases. Determine the price that maximizes the profit of the car park provider if the car park's maximum capacity is 30 cars



Question 9

The demand of a car park in one hour is described by a function D(p) = 30-3p, where D(p) is the number of cars wanted to park at the car park if the price for each parked car is p and the minimum price is p = $1. Basically, the demand decreases as the price increases. Determine the price that maximizes the profit of the car park provider if the car park's maximum capacity is 30 cars

- The profit is calculated from profit multiplied by demand

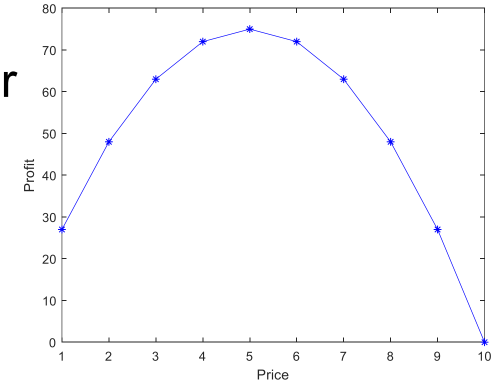
$$p*D(p) = 30p - 3p^2$$

- The price p that maximizes the profit is obtained by differentiating the profit function with respect to the price and equating it to 0,

$$30 - 6*p = 0$$

- Solve for optimal value of price

$$p = 30 / 6 = \$5 \text{ per hour}$$



Question 10

The demands of two car parks in one hour are described by functions D1(p1,p2) = 30-3*p1+p2 and D2(p2,p1) = 60-3*p2+p1, where D1(p1,p2) and D2(p2,p1) the number of cars wanted to park at the car parks 1 and 2, respectively, and p1 and p2 are the prices at the car parks. Basically, the demand of one car park decreases as its price increases and it increases if the price of another car park increases due to competition where users are sensitive to the price of the competitor. Car parks 1 and 2 have the maximum capacities of 30 and 60 cars, respectively.

Determine the prices p1 and p2 that maximize individual profit of each of the car park providers if both the car parks belong to different providers 1 and 2

Question 10

- The profit of car park 1 is
- F1 = p1*D1(p1,p2) = 30*p1 - 3*p1^2 + p1*p2
- The profit of car park 2 is
- F2 = p2*D2(p2,p1) = 60*p2 - 3*p2^2 + p1*p2
- Differentiate the profit of car park 1 with respect to p1
- d F1 / d p1 = 0 = p2 - 6*p1 + 30
- Differentiate the profit of car park 2 with respect to p2
- d F2 / d p2 = 0 = p1 - 6*p2 + 60
- We have two unknown variables and two equations, solve for the optimal value of price p1 and p2 for each of the cark parks

Question 10

- We have two unknown variables and two equations, solve for the optimal value of price p1 and p2 for each of the cark parks
- 6*p1 - 1*p2 = 30 (1)
- -1*p1 + 6*p2 = 60 (2)
- 6*(-1*p1 + 6*p2) = 6*60 multiply (2) with 6
- -6*p1 + 36*p2 = 360
- 35*p2 = 390 add with (1)
- p2 = 390/35 = **$11.1429**
- 36*p1 – 6*p2 = 180 multiply (1) with 6
- 35*p1 = 240 add with (2)
- p1 = 240/35 = **$6.8571**

Question 10

Observation

- p2 = **$11.1429**
- p1 = **$6.8571**
- These prices ensure that "individual profit" of each of the car park owners is "maximized" in the sense that

  there is no other price of one owner that can increase its individual profit if the other owner does not deviate from the above price

- This is the concept of Nash equilibrium for non-cooperative game

Prisoner's Dilemma

- Two prisoners are separated into individual rooms and cannot communicate with each other. Both prisoners understand the nature of the game, have no loyalty to each other.

Prisoner B / Prisoner A	**Prisoner B stays silent** (***cooperates***)	**Prisoner B testifies** (***defects***)
Prisoner A stays silent (***cooperates***)	Each serves 1 year	Prisoner A: 3 years Prisoner B: goes free
Prisoner A testifies (***defects***)	Prisoner A: goes free Prisoner B: 3 years	Each serves 2 years

Strategy from A's Perspective

- If B cooperates
  - If I cooperate, serve 1 year
  - If I defect, go free (better)
- If B defects
  - If I cooperate, serve 3 years
  - If I defect, serve 2 years (better)
- In sum: I should defect regardless of B's choice

- Consequence: both defect (strong Nash equilibrium)

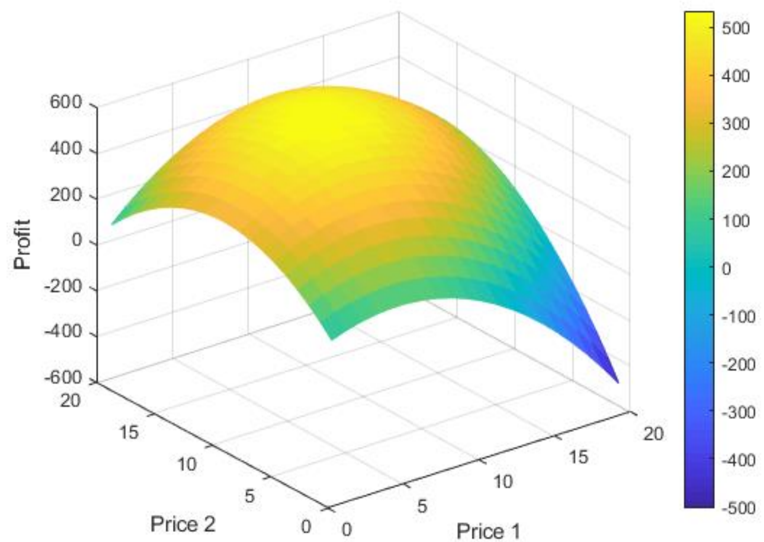
Question 10 (Additional)

The demands of two car parks in one hour are described by functions D1(p1,p2) = 30-3*p1+p2 and D2(p2,p1) = 60-3*p2+p1, where D1(p1,p2) and D2(p2,p1) the number of cars wanted to park at the car parks 1 and 2, respectively, and p1 and p2 are the prices at the car parks. Basically, the demand of one car park decreases as its price increases and it increases if the price of another car park increases. Car parks 1 and 2 has the maximum capacities of 30 and 60 cars, respectively.

Determine the prices p1 and p2 that maximize the profit of the car park provider if both the car parks belong to the same provider

Question 10 (Additional)

- The total profit is
- F = p1*D1(p1,p2) + p2*D2(p2,p1) = 30*p1 - 3*p1^2 + p1*p2 + 60*p2 - 3*p2^2+p1*p2
- Differentiate the profit with respect to p1 and p2
- d F / d p1 = 0 = 2*p2 - 6*p1 + 30
- d F / d p2 = 0 = 2*p1 - 6*p2 + 60
- We have two unknown variables and two equations, solve for the optimal values of prices p1 and p2
- 6*p1 - 2*p2 = 30 (1)
- -2*p1 + 6*p2 = 60 (2)
- 3*(-2*p1 + 6*p2) = 3*60 multiply (2) with 3
- -6*p1 + 18*p2 = 180
- 16*p2 = 210 add with (1)
- p2 = 210/16 = **$13.1250**
- 18*p1 – 6*p2 = 90 multiply (1) with 3
- 16*p1 = 150 add with (2)
- p1 = 150/16 = **$9.3750**



Question 10 (Additional)

Observation

- If both of the car parks belong to the same owner, the price of each car park is higher than that that if the car park belong to different owner (they compete) → should lead to higher profit

Students may check this by themselves